4.1 Rates of change and secants

Definition and results

For $a \neq b$ in the domain of f, the rate of change between a and b is $\frac{f(b)-f(a)}{b-a}$. This is the slope of the secant through the corresponding points. In context, the unit of the rate is the unit of f divided by that of the variable.

Status and justification. Definition; formula for the slope of a line already studied. With $b = a + h$, $h \neq 0$, the rate becomes $\frac{[f(a+h)-f(a)]}{h}$.

Worked example. For $f(x) = x^2$, between 1 and 3, the rate is $\frac{9-1}{3-1}$ = 4. Between 1 and 1 + h, it equals $2 + h$. The slopes for $h = 0{,}1$ and $h = -0{,}1$ are 2.1 and 1.9.

Watch out

The rate is not $\frac{f(b)}{f(a)}$; h is a gap, positive or negative according to the domain.

Immediate application. Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + h.

4.2 Derivative at a point

Definition and results

f is differentiable at a when the rate $\frac{[f(a+h)-f(a)]}{h}$ tends to a real number as h tends to 0, in the domain. This real number is $f'(a)$. An intuitive approach to limits is used here. In kinematics, if f gives a position in metres and time in seconds, the derivative represents an instantaneous speed in m s $^{-1}$.

Status and justification. For $f(x) = x^2$, the rate at a equals 2 $a + h$, which approaches 2 a ; so $f'(a) = 2a$. This calculation will reappear as a derivative function in Chapter 6.

Worked example. For $s(t) = t^2 + 3t$, the rate around 2 is 7 + h, therefore $s'(2) = 7\ \mathrm{m\,s^{-1}}$. The average speed on [2; 3] is 8 m s $^{-1}$; it is not the velocity at time 2.

Watch out

We do not replace h by 0 in an unsimplified quotient: we would get a division by 0.

Immediate application. For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

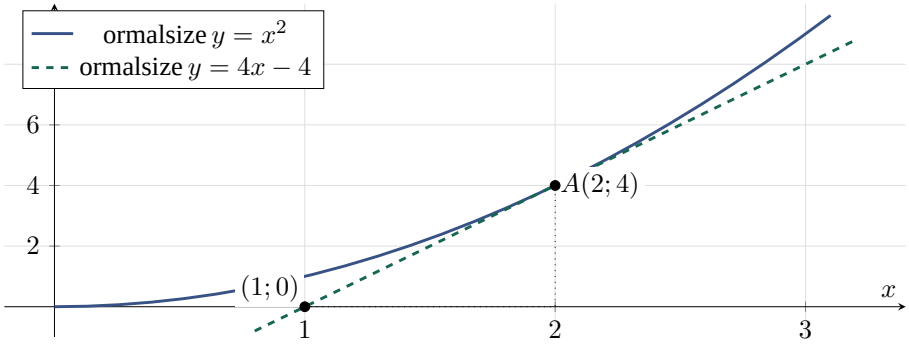
4.3 Tangent and linear approximation

Definition and results

If f is differentiable at a, the tangent at the point ($a; f(a)$) has equation $y = f(a) + f\prime(a)(x - a)$. For small h, we use $f(a + h) \approx f(a) + f\prime(a)h$. This local approximation does not give on its own a bound on the error.

Status and justification. The tangent has slope $f\prime(a)$ and passes through $(a; f(a))$. In $y = f\prime(a)x + p$, substituting this point gives $p = f(a) - af\prime(a)$, hence the equation.

Worked example. For $f(x) = x^2$ at 2, tangent $y = 4 + 4(x - 2) = 4x - 4$. We approximate 2 ,$01^2$ by 4 + 4(0 ,$01) = 4$,04; the exact value is 4 ,0401, gap 0 ,$0001 = h^2$.



The tangent at A also passes through (1; 0). Its slope is $\frac{4-0}{2-1}$ = 4. The dashed drawing lets us distinguish from the curve, even in black and white.

Watch out

A tangent can cross the curve again elsewhere. "A single common point" is not a definition.

Immediate application. We are given $f(3)$ = 5 and $f\prime(3) = -2$. Write the tangent and approach $f(3$,$04)$.

4.4 When differentiability fails

Definition and results

The square root function is defined on [0; + ∞[but is not differentiable at 0. The function absolute value is defined on $\mathbb{R}$ but is not differentiable at 0. Being defined or having a curve continuous therefore does not guarantee differentiability.

Status and justification. For $h > 0$, $\frac{\sqrt{h} - 0}{h} = \frac{1}{\sqrt{h}}$, which does not tend to a finite real number as h approaches 0. For $|x|$, the rate $\frac{|h|}{h}$ is 1 on the right and -1 on the left: no common value.

Worked example. For $|x|$, the secants through the origin all have slope 1 on the right and -1 à left. The corner point does not have a unique tangent with finite slope. For the root, the limiting direction is vertical; there is no real derivative.

Watch out

A very steep gradient is not proof of non-differentiability; we must explain the behaviour of the rate.

Immediate application. Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \overline{x}$ at 0 by its rate of change.